

KERAFLEX

High-performance cementitious adhesive with no vertical slip and extended open time for ceramic and stone material



CLASSIFICATION IN COMPLIANCE WITH ISO 13007-1 AND EN12004

Keraflex is an improved (2) slip resistant (T) cementitious adhesive (C) with extended open time (E) as class C2TE.

WHERE TO USE

Interior and exterior bonding of ceramic tiles porcelain, stone materials, and mosaics of every type on floors, walls, and ceilings. Also suitable for spot bonding of insulating materials such as expanded polystyrene, rock and glass wool, Eraclit® (wood-cement panels), sound-deadening/reduction panels, etc.

Some application examples

- Bonding ceramic tiles, stone materials, and mosaics on the following substrates:
 - cement or mortar wall renders – interior aerated concrete block walls
 - gypsum or anhydrite after having first applied Primer G
 - gypsum board
 - underfloor heating or cooling installations – cement screeds, as long as they are sufficiently cured and dry
 - interior painted walls, provided the paint is completely adhered
 - waterproofing membranes such as **Mapelastick** or **Mapegum WPS**.
- Tile on the tile of existing flooring with tiles up to 900 cm².
- Bonding small-sized ceramic tiles in swimming pools and basins.
- Bonding tiles to floors subject to heavy stress.

TECHNICAL CHARACTERISTICS

Keraflex is a grey or white powder composed of cement and graded aggregates. It contains a high quantity of synthetic resins and special additives created according to a formula developed in MAPEI's research laboratories. A mortar with the following features is obtained when mixed with water:

- easily workable
- highly thixotropic: **Keraflex** can be applied on a vertical surface without sagging or letting the tiles slip, even when heavy tiles are used. Tiles can be installed from the top to the bottom without using spacer pegs
- perfect adherence to all materials normally used in building
- hardens without appreciable shrinkage
- extended open time

RECOMMENDATIONS

Do not use **Keraflex**:

- on the concrete subject to excessive shrinkage
- on walls and floors subject to excessive movement or vibration (wood, fibre-cement, etc.)
- on metal surfaces
- for installations that require a nominal adhesive thickness greater than 5 mm

APPLICATION PROCEDURE

Preparation of the substrate

The substrate must be adequately cured, mechanically sound, free of loose particles, grease, oil, paint, wax, and other deleterious material or surface contamination, and should be sufficiently dry.

Application to damp substrates can slow down the setting of **Keraflex**.

Cementitious substrates must not be subject to shrinkage after the installation. In mild weather, renders must have cured at least 1 week for each centimeter of thickness and cementitious screeds must have cured at least 28 days, unless they have been made with MAPEI special binders for screeds such as **Mapecem**, **Mapecem Pronto**, **Topcem**, **Topcem Pronto**. For surfaces that could become too hot due to exposure to sunlight, consider shading or cooling down with potable water.

Gypsum substrates and anhydrite screeds must be perfectly dry (max. residual moisture 0.5%), sufficiently hard, and free from dust. It is absolutely essential to treat them with **Primer G** or **Mapecrim SP**, while areas subject to extreme dampness must be primed with **Primer S**.

Preparation of the mix

Keraflex must be mixed with clean water to obtain a homogeneous paste free of lumps; leave to rest for approximately 5 minutes and re-mix, the paste is then ready to use. The quantity of water to be used is 30-32 parts per 100 parts (by weight) equal to 7.5-8.0 litres of water for 25 kg of **Keraflex** grey and 36-38 parts per 100 parts (by weight) equal to 9-9.5 litres of water for 25 kg of **Keraflex** white. The mix, produced in this way, is workable for at least 8 hours (at +23°C).

Application of the mix

Keraflex is applied to the substrate using a notched trowel. Choose a trowel that allows complete coverage of the tile backs. To achieve good adhesion, spread an initial thin layer of **Keraflex** on the substrate using the flat side of the trowel. Immediately after, apply the desired thickness of **Keraflex** using a suitable notched trowel, depending on the type and size of the tiles (see "Consumption").

For external ceramic tiled floors and walls, tile sizes greater than 900 cm², floors that must be leveled during tiling installation or subject to heavy loads, when applying in swimming pools and basins filled with water, spread the adhesive on the back of the tile (back-buttering) in order to ensure complete coverage.

Installation of the tiles

It is not necessary to wet the tiles before installation, only in the case of very dusty backs is washing recommended, by quickly immersing them in clean water. The tiles must be installed in the normal way, placing them firmly to ensure good contact with the adhesive. The open time of **Keraflex** in normal conditions of temperature and humidity is 30 minutes; unfavorable environmental conditions (strong sunlight, drying wind, high temperature as well as a highly absorbent substrate), could drastically reduce this time to a few minutes.

Therefore one must constantly check that the adhesive has not formed a superficial skin and is still "fresh", where there is a formation of superficial skin, the adhesive must be reworked with the notched trowel. Wetting the adhesive after it has produced the skin is not recommended because the water forms an anti-adhesive film instead of dissolving it. Tiles can be re-adjusted within 60 minutes (at +23°C) after installation.

Tiles installed with **Keraflex** must not be subject to running water or rain for at least 24 hours and must be protected from frost and strong sunlight for at least 5-7 days after installation.

Spot-bonding insulating materials

For spot-bonding sound-deadening or insulating panels, apply **Keraflex** with a trowel or a float.

GROUTING AND SEALING

Wall joints between the ceramic tiles can be grouted after 4-8 hours and floor joints can be grouted after 24 hours with the specific MAPEI cementitious or epoxy grouts, available in different colours. Expansion joints must be sealed with the specific MAPEI sealants. Grout with the appropriate MAPEI grout (see Technical Data Sheet of grouts for details).

SET TO LIGHT FOOT TRAFFIC

Floors are set to withstand light foot traffic after approx. 24 hours.

READY FOR USE

Surfaces are ready for use after approx. 14 days.
Swimming pools and basins can be filled after 21 days.

CLEANING

Tools and containers should be cleaned with plenty of water while **Keraflex** is still fresh. Surfaces should be cleaned with a damp cloth before the adhesive dries.

CONSUMPTION

Bonding ceramic tiles:

- Mosaics and small sizes in general (trowel n. 4): 2 kg/m²
- Normal sizes (trowel n. 5): 2.5-3 kg/m²
- Large sizes, exterior floors (trowel n. 6): 4 kg/m²

Spot-bonding insulating material:

- Foam, etc. approx. 0.5-0.8 kg/m²
- Gypsum board panels, cellular concrete: approx. 1.5 kg/m²

PACKAGING

Keraflex white and grey are available in 25 kg bags.

STORAGE

When stored in dry conditions in the original, unopened bags, **Keraflex** has a shelf life of 12 months. If stored at high temperature and or high humidity conditions the shelf life may be reduced.



Laying polystyrene foam slabs with Kera



Installation of pre-glossed marble on floors



Installation of single-fired tile on expanded foam concrete block walls



Single-fired tile on terrazzo tile of an external wall





Tile on tile with **Keraflex** grey

SAFETY INSTRUCTIONS FOR PREPARATION AND APPLICATION

Instruction on the safe use of our products can be found on the latest version of the Safety Data Sheet available on our website www.mapei.ae.

TECHNICAL DATA (typical values) In compliance with: – European EN 12004 such as C2TE – ISO 13007-1 such as C2TE – American ANSI A 118.4	
PRODUCT IDENTIFICATION	
Type:	powder
Colour:	white or grey
Bulk density (kg/m ³):	1,300
Dry solids content (%):	100
EMICODE:	EC1 Plus - very low emission
COMPOSITION AND PROPERTIES OF THE MIXTURE (at +23°C and 50% R.H.)	
Mixing Ratio:	100 parts Keraflex with 30-32 parts water by weight for Keraflex grey and 36-38 parts water by weight for Keraflex white
Consistency of the mix:	paste
Density of the mix (kg/m ³):	1500
pH of mix:	13
Pot life:	over 8 hours
Application temperature range:	from +5°C to +40°C
Open time (acc. EN 1346):	> 30 minutes
Adjustment time:	approx. 60 minutes
Ready for grouting on walls:	after 4-8 hours
Ready for grouting on floors:	after 24 hours
Set to light foot traffic:	24 hours

Ready for use:	14 days
FINAL PERFORMANCE	
Bonding strength according to EN 1348 (N/mm ²):	
– initial bonding after 28 days:	1.8
– bonding after heat exposure:	1.7
– bonding after immersion in water:	1.2
– bonding after freeze/thaw cycles	1.4
Resistance to alkalis:	excellent
Resistance to oils:	excellent (poor to vegetable oils)
Resistance to solvents:	excellent
In-service temperature:	from –30°C to +90°C



An example of an installation of porcelain tiles - Zanchetta Shopping Centre - Treviso (Italy)

IMPORTANT NOTES

Whilst we try to ensure that any advice, recommendations or information given in our literature is accurate and correct, we have no control over the circumstances in which our product is used. It is therefore important that installers satisfy themselves that the product and conditions are suitable for the envisaged application. No warranty can be given or responsibility accepted other than, that the product supplied by us will meet our written specification. The installer should ensure that our latest product data and safety information sheets have been consulted prior to use.

Please refer to the current version of the Technical Data Sheet, available from our website www.mapei.com

LEGAL NOTICE

The contents of this Technical Data Sheet ("TDS") may be copied into other project-related documents, but the resulting document shall not supplement or replace requirements per the TDS in force at the time of the MAPEI product installation. The most up-to-date TDS can be downloaded from our website www.mapei.com.

ANY ALTERATIONS TO THE WORDING OR REQUIREMENTS CONTAINED OR DERIVED FROM THIS TDS EXCLUDES THE RESPONSIBILITY OF MAPEI.

All relevant references for the product are available upon request and from www.mapei.com.

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